

2022

Bachelor in Computer Applications

Course Title: **Mathematics II**

Code No: **CACS154** Semester: **II**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

Group B

2. "A function $f(x)$ is defined as
$$f(x) = \begin{cases} 2x + 3 & \text{for } -\frac{3}{2} \leq x < 0 \\ 3 - 2x & \text{for } 0 \leq x \leq \frac{3}{2} \\ -3 - 2x & \text{for } x > \frac{3}{2} \end{cases}$$
 show that $f(x)$ is continuous at $(x = 0)$ and discontinuous at $(x = \frac{3}{2})$."
3. Find $\frac{dy}{dx}$ when $(x = a(t + \sin t))$, $(y = a(1 - \cos t))$.
4. State L'Hospital's Rule. Use it to evaluate: $\lim_{x \rightarrow 0} \frac{e^x - \ln(1+x)}{x^2}$
5. State Rolle's Theorem. Verify Rolle's Theorem for the function $(f(x) = x^2 - 9)$ in the interval $(-3 \leq x \leq 3)$.
6. Evaluate: a) $\int_0^2 \frac{dx}{x^2 - 25}$ b) $\int \frac{2x+5}{\sqrt{x^2}} + 5x \, dx$
7. Solve: $(\sin x \frac{dy}{dx} + (\cos x) y = -\sin x \cos x)$.
8. Solve the following system by Gauss-Seidel method:
$$\begin{cases} -4x + y - z = -8 \\ 3x + 6y + 2z = 1 \\ x - y + 3z = 2 \end{cases}$$

Group C

Attempt any TWO questions [2x10=20]

9. Using simplex method, find optimal solution of $(Z = 7x_1 + 5x_2)$ Subject to $(x_1 + 2x_2 \leq 6)$ $(4x_1 + 3x_2 \leq 6)$ $(x_1 \geq 0, x_2 \geq 0)$.
10. What do you mean by stationary points and inflection points? Using derivatives, find two numbers whose sum is 20 and sum of whose squares is minimum.
11. Using Simpson's $\frac{1}{3}$ rule evaluate $\int_0^1 \frac{1}{1+x} \, dx$ with 3 points of intervals. Find error of approximation. How many points are to be considered to make the approximation value within 10^{-5} ?